AIAA Atmospheric Flight Mechanics Conference
10 - 13 August 2009, Chicago, Illinois

AIAA 2009-5941

Check for updates

Aircraft Parameter Estimation using Neural Network based Algorithm

N. K. Peyada* and A. K. Ghosh†
Department of aerospace Engineering
Indian Institute of Technology Kanpur, UP-208016, INDIA

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

Aircraft Parameter estimation is probably the most outstanding and illustrated example of the system identification methodology. In the past the most widely used parameter estimation methods have been Equation error method, Output error method, Maximum likelihood method and Filter error method. In this paper an algorithm based on neural modeling and Gauss-Newton optimization is proposed to estimate aerodynamic parameters from flight data. The proposed method bypasses the requirement of solving the equations of motion to predict motion variables. The proposed algorithm uses a black box approach to build the flight dynamic model of an aircraft using measured flight variables. The algorithm was initially validated on flight data generated using HANSA-3 aircraft at Indian Institute of Technology Kanpur. Similar validation was carried out with lateral-directional flight data of ATTAS aircraft (supplied by DLR Germany). The estimated values were compared with estimates obtained using Filter error method and Least square method. The applicability of the proposed algorithm in handling flight data with atmospheric turbulence was also investigated by applying it on real flight data of test aircraft, HFB-320. The work concludes by presenting a comparison among the estimates obtained by applying several methods on real flight data of three different test aircraft.

Nomenclature

a_x, a_y, a_z	=	accelerations component along x, y, z body axes
b	=	span length
$\bar{c}$	=	mean aerodynamic cord length
C_ℓ, C_m, C_n	=	rolling, pitching and yawing moment coefficient
C_D, C_L	=	drag and lift coefficient
C_X, C_Y, C_Z	=	force coefficient along x, y, z body axes
e^j	=	column vector with one in the jth row and zeros elsewhere
F_{eng}	=	thrust force by aircraft engine
I_x, I_y, I_z	=	moments of inertia about x, y and z axes
I_{xz}	=	moment of inertia about xz plane
m	=	mass of the aircraft
p, q, r	=	roll, pitch and yaw rates
$\dot{p}, \dot{q}, \dot{r}$	=	roll, pitch and yaw accelerations
$\bar{q}$	=	dynamic pressure
S	=	reference wing area
$U(k)$	=	input vector at k^{th} instant

* Ph.D. Student, Department of Aerospace Engineering , nkpeyada@gmail.com
† Professor, Department of Aerospace Engineering, akg@iitk.ac.in, Member AIAA

Copyright © 2009 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved.

V = true airspeed
W = weight matrix in neural network
$Y(k)$ = Neural network out put vector at k^{th} instant
$Z(k)$ = measured output vector at k^{th} instant

Greek Notation

α = angle-of-attack
β = angle of side-slip
ϕ, θ, ψ = roll, pitch and yaw angles
Θ = system parameters
$\delta_a, \delta_e, \delta_r, \delta_f$ = aileron, elevator, rudder and landing-flap deflections
σ_{eng} = engine inclination angle with aircraft body axes

Abbreviation

CG = center of gravity
ENCG = distance between center of gravity and engine

I. Introduction

AIRCRAFT system identification is primarily concerned with providing a mathematical description for the aerodynamic forces and moments in terms of relevant measurable quantities such as control surface deflection, aircraft angular velocity, airspeed or Mach number, and orientation of the aircraft to the relative wind. Aerodynamic parameters quantify the functional dependence of the aerodynamic forces and moments on measurable quantities, when the mathematical model is parametric.[1] The parameter estimation process consists of finding values of unknown model parameters in an assumed model structure, based on noisy measurements. Estimation of aerodynamic parameters could easily be done by several methods; however, complicating issues arise when considering application to real systems. The first complication arises due to imperfect measurements of motion variables. In practical applications sensor errors are unavoidable. The second complication of real system is the presence of state/process noise.[2] The final complication of real system is the absence of exact mathematical model of the system under investigation. Most of the conventional methods assume that the system is correctly described by the dynamic model. Physical systems are seldom described exactly by dynamic models, so the question of modeling error arises. There is no comprehensive theory of modeling error available. Any modeling error is simply treated as state and/or measurement noise, in spite of the fact that the modeling error may be deterministic rather than random. The assumed noise statistics can then be adjusted to include the contribution of the modeling error.[1-3] In the past the most widely used parameter estimation methods have been the output error (OE) method, maximum likelihood (ML) method and Filter error method (FEM).[3-5]

For more than three decades, maximum likelihood parameter estimators have been successfully applied for extraction of aircraft stability and control derivatives (parameters) from flight data. The application of ML estimators to flight data with either measurement noise or process noise has been accepted as a standard approach for parameter estimation. However, in presence of both measurement and process noise, the ML estimator might lead to convergence problems and other practical difficulties.[3-5] The Filter error method,[6-8] has been shown capable of handling flight data with measurement and process noise, and also its extension for nonlinear systems has been proposed and validated by Jategaonkar and Plaetschke.[9] The filter error method faces difficulties in extending it to multiple experiments treating process noise distribution matrix separately for each maneuver. Further, this method always yields a good match for the responses which do not necessarily guarantee correct postulation of the model.[3] It is for these reasons that the filter error method is used in practice for special cases only. Most of the practical problems in parameter estimations are dealt by output error or least squares methods.

The least square (LS) techniques, also called regression analysis, belong to a class of methods called the equation error method which does not require explicit postulation of flight dynamic model. Equation error method directly defines a cost function in terms of an input-output equation. The cost function is not based on probability theory. The linear least square technique is characterized by its mathematical simplicity. However, the performance and applicability of LS method is strongly influenced by the data quality, assumption made regarding the noise and availability of the necessary variables. Applications of least square technique do not require postulation of mathematical model of the system under investigation.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

Recently neural network based methods have been proposed to estimate aerodynamic parameters from flight data.[10,11] Such approach generates a model based entirely on the input/output measurements of the system without trying to model the internal physical mechanism of the system. Since we wish to estimate and validate such models using measured input/output data, the Artificial Neural Networks (ANNs) provide an alternative approach to model building. It is important to note that such an approach of model building for complex systems neither require postulation of mathematical model nor solving of equations of motion. More recently, many scientists and engineers have explored the potential of ANNs in diverse fields such as signal processing, pattern recognition, aircraft aerodynamic modeling, parameter estimation and control.[12] Artificial neural networks have been used in variety of applications because they are adaptive, they learn through examples and they can provide excellent function approximation.[12,13] Artificial neural networks have been used to model aircraft dynamics where aircraft motion and control variables are mapped to predict the total aerodynamic coefficients.[14-16] In all these papers, the emphasis has been on aerodynamic modeling and estimation of aerodynamic coefficients using Feed Forward Neural Networks (FFNNs). Raisinghani, Ghosh et al.[10,11] using FFNNs proposed two new methods namely the Delta and the Zero method for explicitly estimating aircraft parameters from flight data. The Delta method and the Zero method based on linear regression may apparently look similar.[3] However, the advantage of FFNNs in handling noise (having zero mean) in flight data to yield better and consistent estimates can not be ignored. Further, the neural based methods have the advantage of being able to use a true non-linear function as the working model for prediction.

Recently, an algorithm using FFNNs and Gauss-Newton optimization method to estimate aerodynamic parameters from flight data of aircraft has been proposed in Ref. 17. The algorithm starts with establishing neural model using time histories of motion and control variables of aircraft in flight. Once this model is validated, it can be used to compute response for any arbitrary control input. However, the proposed neural model does not represent a generic flight dynamic model. The proposed neural model can only be used to predict time histories of motion variables at $(k+1)^{th}$ instant given the measured initial conditions corresponding to k^{th} instant (where k=1 to n; n is the total no. of discrete data point). Similarly for computing time histories at $(k+2)^{th}$ instant, the neural model uses measured initial conditions corresponding to $(k+1)^{th}$ instant. For all practical purposes of parameter estimation, this approach helps in building flight dynamic model using measured input-output data and does not require any priori postulation of mathematical model or solving of equations of motion. Since the neural mapping uses measured motion variables, the performance and applicability of the proposed technique is also influenced by data quality. Special care needs to be taken in selecting tuning parameters, to avoid overtraining. Specially selection of number of iteration and number of neuron in hidden layer plays important role during NN modeling while handling flight data with noise. In general, it is not necessary that the larger, the number of training iterations,[3] is, the better the trained FFNN, because the network starts to fit the noise as well. For optimization using Gauss-Newton method, such trained neural model can conveniently be used to compute system state and responses, error cost function and response gradient.[17]

This algorithm was applied on real flight data pertaining to flight of three different aircraft. Since the neural model has capability to model flight data having atmospheric turbulence,[3] attempt has been made to evaluate performance of the proposed algorithm in handling real flight data with atmospheric turbulence.

The verification of the identified model is a key step in the identification process to assess the predictive capabilities of the extracted model. Proof of match exercise has been carried out to validate the estimated model. It is a widely used approach which is based on the comparison of model prediction with the flight measurements. To this end, flight data omitted from the identification studies is selected to ensure that the model is not tuned to specific data record or input form. The effect of choice of magnitude and sign of the initial guess values of the parameters on the estimates has also been studied. Various test runs were carried out with different sets of initial guess values of the parameters. It was observed that the proposed algorithm has the capability of extracting the aerodynamic parameters accurately even if the initial guess values were substantially off (both in sign and magnitude) from nominal values of the aerodynamic parameters of the chosen aircraft.

II. Parameter Estimation Algorithm

The estimation process starts with conducting of experiments. Predecided maneuvers are executed to excite selected dynamics of the chosen aircraft. On board data acquisition system is activated to acquire the flight data. These sets of flight data are referred to as measured flight data. The measured flight data undergo data compatibility check and all the variables are transferred to center of gravity (cg) of the aircraft for further analysis.

Let the measured flight data at the kth instant contain the time histories of angle of attack $\alpha(k)$, pitch angle $\theta(k)$, pitch rate $q(k)$, airspeed $V(k)$, accelerations $a_x(k)$ and $a_z(k)$. Construction of appropriate input

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

and output vectors using measured flight data is one of the important steps to build the neural model. Let the input vector $U(k)$ be constructed as per Eq. (1).

$$U(k)=\left[\alpha(k),\theta(k),q(k),V(k),C_D(k),C_L(k),C_m(k)\right] \tag{1}$$

Where the values of $C_D(k),C_L(k)$ and $C_m(k)$ at the k[th] instant are obtained by plugging measurement variables $a_x(k),a_z(k)$ and dynamic pressure $\bar{q}(k)$ along with mass and inertia characteristics in Eqs. (2)-(4).

$$C_D(k)=-C_X(k)\cos\alpha(k)-C_Z(k)\sin\alpha(k) \tag{2}$$

$$C_L(k)=C_X(k)\sin\alpha(k)-C_Z(k)\cos\alpha(k) \tag{3}$$

$$\begin{aligned}C_m(k)=&[I_y\dot{q}(k)-I_{xz}\left(p^2(k)-r^2(k)\right)-(I_z(k)-I_x(k))p(k)r(k)\\&-F_{eng}\cos\sigma_{eng}Z_{ENCG}-F_{eng}\sin\sigma_{eng}Y_{ENCG}]/\left(\bar{q}(k)S\bar{c}\right)\end{aligned} \tag{4}$$

Where $C_X(k)$ and $C_Z(k)$ are computed using Eqs. (5)&(6).

$$C_X(k)=ma_x^{CG}/(\bar{q}S) \tag{5}$$

$$C_Z(k)=ma_z^{CG}/(\bar{q}S) \tag{6}$$

And p,q,r are the angular rates, σ is the engine setting angle, Z_{ENCG},Y_{ENCG} are the offset distance of engine from CG and F_{eng} is the engine thrust.

Let the output vector $Z(k+1)$ at $(k+1)^{th}$ instant required for building neural model be constructed as per Eq. (7).

$$Z(k+1)=\left[\alpha(k+1)\ \ \theta(k+1)\ \ q(k+1)\ \ V(k+1)\ \ a_x(k+1)\ \ a_z(k+1)\right]^T \tag{7}$$

A typical FFNN structure for developing neural model to be used for parameter estimation is presented in Fig. 1. Once the input vector $U(k)$ and the output vector $Z(k+1)$ are constructed using flight data, the next task is to develop the neural model using FFNN (Fig. 1). The neural network performance, that is its ability to accurately duplicate data used in training with adequate prediction capability, depends on the network tuning parameters.[12] The choice of tuning parameters depends on several factors, such as number of inputs and outputs, the amount of noise in data, and the complexity of the input-output subspace. Several guidelines to tune the FFNN are available in open literature.[12] Once neural model is created, it can be used for prediction of output variable $Y(k+1)$ for any given measured input variable $U(k)$ at k[th] instant.

For parameter estimation the input vector $U(k)$ is reconstructed by keeping the same initial conditions $(\alpha(k),\theta(k),q(k),V(k))$ used for training however, $C_D(k),C_L(k)$ and $C_m(k)$ corresponding to identical control input used for generating flight data are modified as per the chosen aerodynamic model in the estimating algorithm. Let say, the chosen aerodynamic model for longitudinal parameter estimation be as given in Eqs. (8), (9) and (10).

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

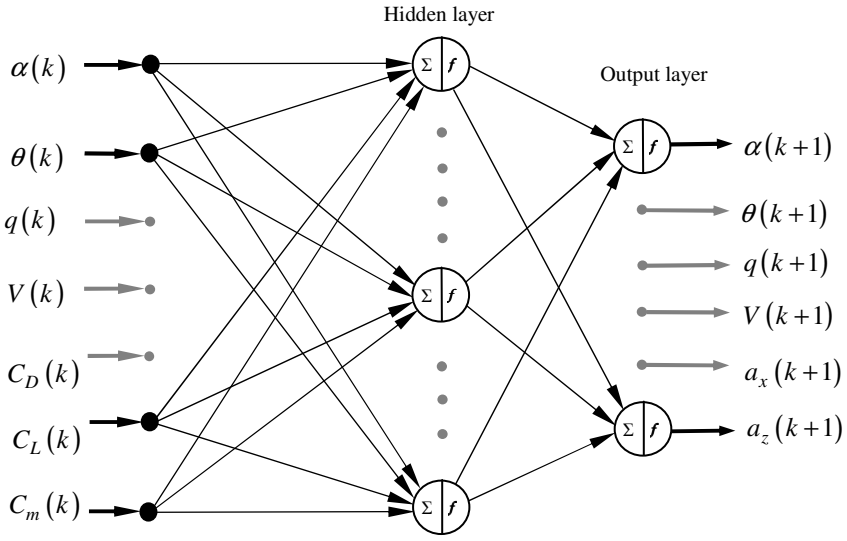


Figure 1. Neural-Architecture for Flight Dynamic Model.

$$C_D = C_{D_0} + C_{D_\alpha}\alpha + C_{D_{\delta_e}}\delta_e \tag{8}$$

$$C_L = C_{L_0} + C_{L_\alpha}\alpha + C_{L_q}\left(\bar{q}\,\bar{c}/2V\right) + C_{L_{\delta_e}}\delta_e \tag{9}$$

$$C_m^{CG} = C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\left(\bar{q}\,\bar{c}/2V\right) + C_{m_{\delta_e}}\delta_e \tag{10}$$

The aim of the exercise is to estimate unknown parameter vector Θ consisting of aerodynamic parameters namely, C_{D_0}, C_{D_α}.... C_{m_q} and $C_{m_{\delta_e}}$ as given in Eq. (11).

$$\Theta = \left[\, C_{D_0}\ C_{D_\alpha}\ C_{D_{\delta_e}}\ C_{L_0}\ C_{L_\alpha}\ C_{L_q}\ C_{L_{\delta_e}}\ C_{m_0}\ C_{m_\alpha}\ C_{m_q}\ C_{m_{\delta_e}} \,\right]^T \tag{11}$$

To start the estimation algorithm it is necessary to specify some suitable initial guess values of the unknown parameters vector Θ, consisting of non-dimensional parameters used for the description of aerodynamic model of C_D, C_L and C_m^{CG}. Using the initial guess values of the aerodynamic parameters the estimated coefficients C_{De}, C_{Le} and C_{me}^{CG} are computed using Eqs.(8)-(10). Next, the estimated response of the aircraft $y(k+1)$ at (k+1)th instant is computed using input vector $U(k)$ having same measured flight data (as given in Eq. (7)) with C_D, C_L and C_m^{CG} replaced by C_{De}, C_{Le} and C_{me}^{CG}). Let this estimated response be represented by output vector as given in Eq. (12).

$$Y(k+1) = \left[\, \alpha(k+1)\ \ \theta(k+1)\ \ q(k+1)\ \ V(k+1)\ \ a_x(k+1)\ \ a_z(k+1) \,\right]^T \tag{12}$$

The residual error $E(k)$ and covariance matrix of the residual R are computed using Eqs. (13) and (14). Next the value of $E(k)$ and R are used to compute error cost function $J(\Theta, R)$ using Eq. (15).

$$E(k) = \left[\, Z(k) - Y(k) \,\right] \tag{13}$$

$$R = \frac{1}{N}\sum_{k=1}^{N}\left[Z(k) - Y(k)\right]\left[Z(k) - Y(k)\right]^T \tag{14}$$

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

$$J(\Theta,R)=\tfrac{1}{2}\sum_{k=1}^{N}\left[Z(k)-Y(k)\right]^{T}R^{-1}\left[Z(k)-Y(k)\right] \tag{15}$$

Finally, the Gauss-Newton method has been applied to update the parameter vector Θ by minimizing the error cost function $J(\Theta,R)$ through Eqs. (16) to (19).

$$\Theta_{i+1}=\Theta_i+\Delta\Theta, \quad \text{and} \quad \mathcal{F}\Delta\Theta=-\mathcal{G} \tag{16}$$

$$\mathcal{F}=\sum_{k=1}^{N}\left[\frac{\partial Y(k)}{\partial\Theta}\right]^{T}R^{-1}\left[\frac{\partial Y(k)}{\partial\Theta}\right] \tag{17}$$

$$\mathcal{G}=-\sum_{k=1}^{N}\left[\frac{\partial Y(k)}{\partial\Theta}\right]^{T}R^{-1}\left[Z(k)-Y(k)\right] \tag{18}$$

$$\left[\frac{\partial Y(k)}{\partial\Theta}\right]_{ij}=\frac{Y_{pi}(k)-Y_i(k)}{\partial\Theta_j} \tag{19}$$

The sensitivity matrix $\left[\frac{\partial Y(k)}{\partial\Theta}\right]_{ij}$ at each iteration (i) is computed using approximate relation given by Eq. (19). The numerical values of Y_{pi} (perturbed response) correspond to perturbed parameters $\Theta+\Delta\Theta$ represented in Eq. (16) are obtained using already trained neural model. The estimation error covariance matrix *P* is function of model parameters Θ, the data points being analyzed and the covariance matrix of the residuals *R*. The expression to compute *P* is given in Eq. (20).

$$P\approx\left\{\sum_{k=1}^{N}\left[\frac{\partial Y(k)}{\partial\Theta}\right]^{T}R^{-1}\left[\frac{\partial Y(k)}{\partial\Theta}\right]\right\}^{-1} \tag{20}$$

The standard deviations of the parameter estimates are the square root of diagonal elements of the estimation error covariance matrix *P* which can be computed using Eq. (21).

$$\sigma_{\Theta i}=\sqrt{P_{ii}} \tag{21}$$

III. Flight Data of Different Aircrafts

A. Longitudinal Flight Data of HANSA-3 Aircraft

The final test of any scheme for parameter estimation must come from its successful demonstration on real flight data. A flight data base for identification studies was gathered from flight maneuvers with the test aircraft.[18] Typically, starting from trim flight conditions, the pilot applied control input in an attempt to excite the chosen dynamic modes. An onboard measurement system installed on the test aircraft provided measurements using dedicated sensors of a large number of signals such as aircraft motion variables, atmospheric conditions, control surface position etc. The measurements made in flight were recorded (sampling rate 50 Hz) onboard using suitable interface with standard laptop. The flight data had raw data for measured $V,\alpha,\beta,p,q,r,a_x,a_y,a_z,\phi,\theta,\psi,H,\delta_e,\delta_a$ and δ_r and the location of the measuring sensors. The measurement of airspeed (V), angle of attack (α), angle of side slip (β) were obtained with flight log mounted on a boom fixed to the tip of the wing. The angular rates p,q and r were obtained from the measurements available from the inertial platform. The accelerations along the three body axes were measured using an accelerometer triad located near the CG of the aircraft. The angular rates $\dot{p},\dot{q}$ and $\dot{r}$ were obtained by numerical differentiation of the corresponding angular rates. The control surface deflections $(\delta_e,\delta_a,\delta_r)$ were measured using potentiometer. The temperature

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

T was recorded using the standard cockpit outside air temperature (OAT) gauge. Two sets of flight data simulating short period longitudinal dynamics were generated at an altitude 2000 m. The cruise speed at which the perturbations were initiated was fixed at nearly 46 m/s.

The longitudinal flight data (FLT1) was generated using multi step elevator input $\left(\delta_{e_{\max}} = 7\,\text{deg}\right)$ having total duration of 3s only. Another flight data set (FLT2) was generated with two similar looking double pulse input having almost same magnitude. These two pulses although look similar but have opposite elevator deflections to excite the longitudinal dynamics. The flight data FLT1 and FLT2 are used for parameter estimation and proof-of-match exercise. Figure 2 typically presents flight data FLT1 for ready reference.

B. Lateral Directional Flight Data of ATTAS Aircraft

The effectiveness of the proposed method for estimation of lateral-directional aerodynamic parameters was investigated using real flight data supplied by DLR Germany for the ATTAS aircraft.[3] The flight data for the flight condition defined by a landing-flap condition δ_f = special position (1 deg) and perturbation about steady state flight at indicated airspeed (IAS), V_{IAS} = 250 knots were analyzed. Unlike longitudinal parameter estimation, the lateral directional parameter estimation used at least two control inputs, namely aileron $\left(\delta_a\right)$ and rudder $\left(\delta_r\right)$. It is well established that a disjointed control input maneuver (with δ_a and δ_r not occurring simultaneously) would result in better estimation of overall lateral-directional parameters.[3,7,10] In view of the above, flight data FLT3 and FLT4 pertaining to lateral-directional dynamics were chosen for parameter estimation. The flight data FLT3 is presented in Fig. 3.

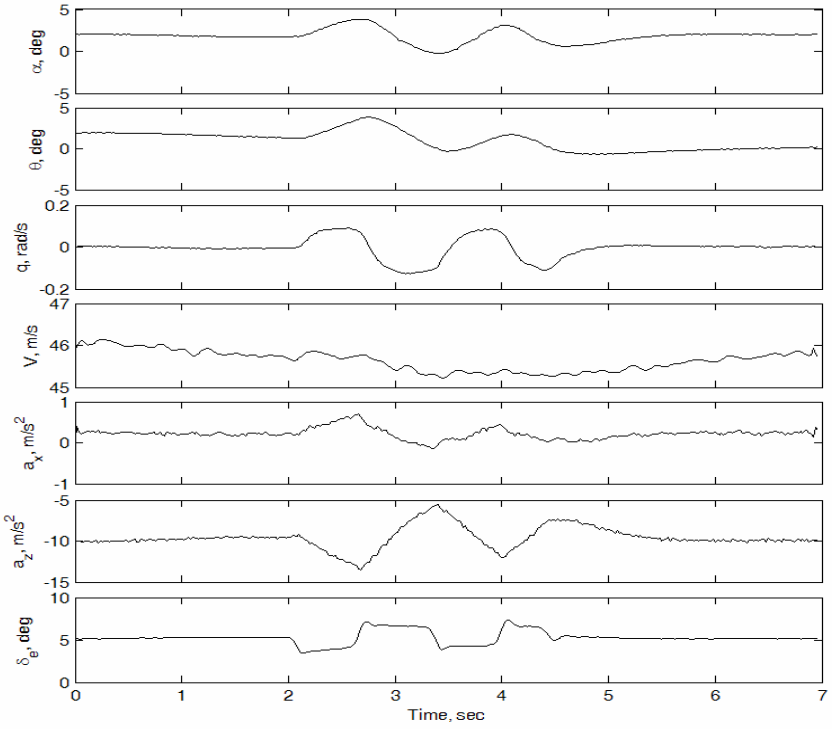


Figure 2. Longitudinal Flight Data: FLT1.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

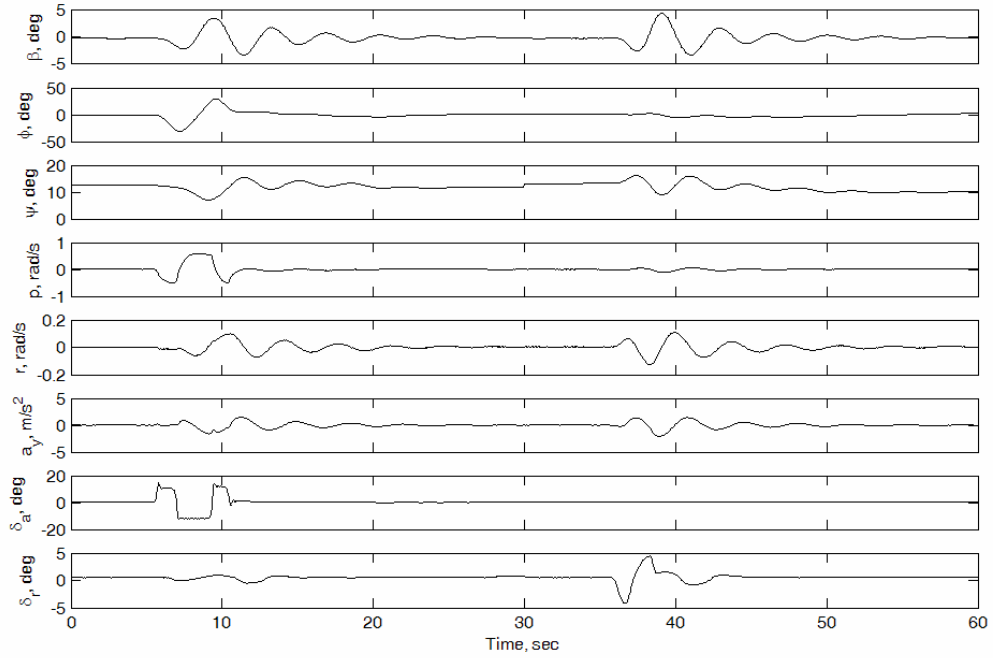


Figure 3. Lateral-Directional Flight Data: FLT3.

C. Flight Data of HFB-320: Longitudinal Mode with Atmospheric Turbulence

The proposed method was applied on flight data having process noise. Flight test results reported in Ref. 3 were used for this purpose. Flight tests were carried out using research aircraft HFB-320 to excide the longitudinal motion through a multi step elevator input.

IV. Parameter Estimation

The proposed method uses neural mapping for flight dynamic modeling and Gauss-Newton optimization technique for aerodynamic parameters updation during iteratively minimizing the error cost function. The proposed method is nomenclatured as Neural-Gauss-Newton (NGN) method for further reference. Five sets of flight data were processed using the NGN, FEM and LS methods. The subsequence section presents the results obtained using the above methods.

A. Parameter Estimated from Longitudinal Flight Data of HANSA-3 Aircraft using NGN, FEM, LS Method

The flight data FLT1 and FLT2 were generated by exciting the longitudinal dynamics of HANSA-3 test aircraft. The longitudinal short period dynamics was excited about a steady state trim at V = 46 m/s at a cruise altitude of 2000 m. Day and time of the experiment were carefully chosen to ensure fairly calm weather. The flight data acquired were preprocessed and reconstructed after carrying out exhaustive data compatibly check. Next, using these flight data, the input and output vectors for neural mapping were constructed. The input vector $U(k)$ consisted of time histories at k^{th} instant of $\alpha(k), \theta(k), q(k), V(k), C_D(k), C_L(k)$ and $C_m(k)$ whereas the output vector $Z(k+1)$ had $\alpha(k+1), \theta(k+1), q(k+1), V(k+1), a_x(k+1)$ and $a_z(k+1)$ as its elements. Using $U(k)$ as network input vector and $Z(k)$ as the output vector, the FFNN was trained as discussed earlier. The FFNN used a log-sigmoid and linear transfer function as the activation function and Levenberg-Marquardt,[15,16] algorithm was used for updating the neural network weights. The mean square error (MSE) criterion or the number of iteration decided the termination of the iterative process. A range of values of the network parameters was tried to arrive at the final architecture of the FFNN used for parameter estimation. The network parameters varied were the number of hidden layers (1 - 3), the number of hidden neurons in the hidden layers (2-15), the learning rate (0.1 - 0.8), the momentum rate (0.1 – 0.8), and the number of iterations (100 – 2000). The network parameters finally chosen gave a good

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

match between the true and the predicted values of the time histories of the variables. The final FFNN used had one hidden layer with ten neurons, the learning rate = 0.3.

The NGN, FEM and LS methods have then been used to predict the stability and control derivatives (parameters) of the aircraft. The values of the estimated parameters namely $C_{D_0}, C_{D_\alpha}, C_{D_{\delta_e}}, C_{L_0}, C_{L_\alpha}, C_{L_{\delta_e}}, C_{m_0}, C_{m_\alpha}, C_{m_q}$ and $C_{m_{\delta_e}}$ along with their Cramer-Rao bounds are presented in Table 1. Column 2, 3 and 4 of Table 1 present the values of the estimated parameters obtained by applying NGN, FEM and LS method respectively on flight data FLT1. Similarly, column 5, 6 and 7 represent estimated parameters obtained by applying NGN, FEM and LS method on flight data FLT2. It could be seen in Table 1 that the values of the parameters obtained using the NGN method are in fairly close agreement with the values obtained using FEM and LS method. It may be mentioned here that despite the best effort of the pilot, it was not possible to generate flight data pertaining to 3-2-1-1 type elevator input excitation 3. This could possibly be one of the reasons for observed scatter in the estimated values of the weak derivatives namely $C_{L_{\delta_e}}$.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

Table 1. HANSA-3 Longitudinal Parameters

Parameter	FLT1			FLT2		
	NGN	FEM	LS	NGN	FEM	LS
C_{D_0}	0.0343 (0.0006)*	0.0343 (0.00139)	0.0349 (0.00094)	0.0353 (0.00069)	0.0352 (0.00116)	0.0359 (0.00089)
C_{D_α}	0.1991 (0.00598)	0.1952 (0.01363)	0.1827 (0.00942)	0.2388 (0.00606)	0.2489 (0.01035)	0.2311 (0.00789)
$C_{D_{\delta_e}}$	0.181 (0.00632)	0.1798 (0.01493)	0.1804 (0.00997)	0.167 (0.00703)	0.1598 (0.01202)	0.1626 (0.00908)
C_{L_0}	0.3755 (0.00238)	0.3562 (0.01051)	0.3592 (0.00395)	0.3535 (0.00314)	0.3288 (0.00747)	0.3414 (0.00473)
C_{L_α}	4.87 (0.02638)	4.9692 (0.11522)	4.6329 (0.04705)	4.9719 (0.02927)	4.994 (0.07376)	4.891 (0.04747)
C_{L_q}	21.302 (0.72701)	21.45 (2.9955)	26.481 (1.22295)	18.02 (0.88719)	19.421 (2.1665)	19.494 (1.4193)
$C_{L_{\delta_e}}$	0.3083 (0.03066)	0.4647 (0.13291)	0.5716 (0.05138)	0.2605 (0.03808)	0.5051 (0.09037)	0.4272 (0.05781)
C_{m_0}	0.1073 (0.00266)	0.1021 (0.00282)	0.0999 (0.00318)	0.1154 (0.00351)	0.1039 (0.00223)	0.1036 (0.00284)
C_{m_α}	-0.4969 (0.03717)	-0.4878 (0.03141)	-0.5304 (0.03786)	-0.4275 (0.03548)	-0.4154 (0.02168)	-0.4375 (0.02847)
C_{m_q}	-6.5781 (0.98897)	-4.7902 (0.83936)	-3.704 (0.98427)	-6.5309 (1.1446)	-6.0379 (0.66239)	-5.6789 (0.85133)
$C_{m_{\delta_e}}$	-0.9304 (0.03535)	-0.877 (0.03589)	-0.8364 (0.04136)	-1.0121 (0.0414)	-0.9039 (0.0271)	-0.8887 (0.03467)

* *Values in parentheses indicate sample standard deviation.*

Next, we carried out the method verification exercise using flight data. Both the flight data FLT1 and FLT2 were used for parameter estimation and FLT2 was used for verification using proof-of-match procedure.[3] In the first step, the estimated aerodynamic parameters obtained through flight data FLT1 were used in solving longitudinal equations of motion.[3] The flight responses were generated corresponding to initial conditions and elevator input used for given flight data FLT2. The flight responses computed using estimated parameters obtained using NGN, FEM and LS method are referred to as Est-NGN, Est-FEM and Est-LS respectively in the text. A comparison of the motion variables is obtained with the measured flight data FLT2 and has been presented in Fig. 4. A fairly close match among the responses generated using NGN, FEM and LS estimates were observed in Fig. 4.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

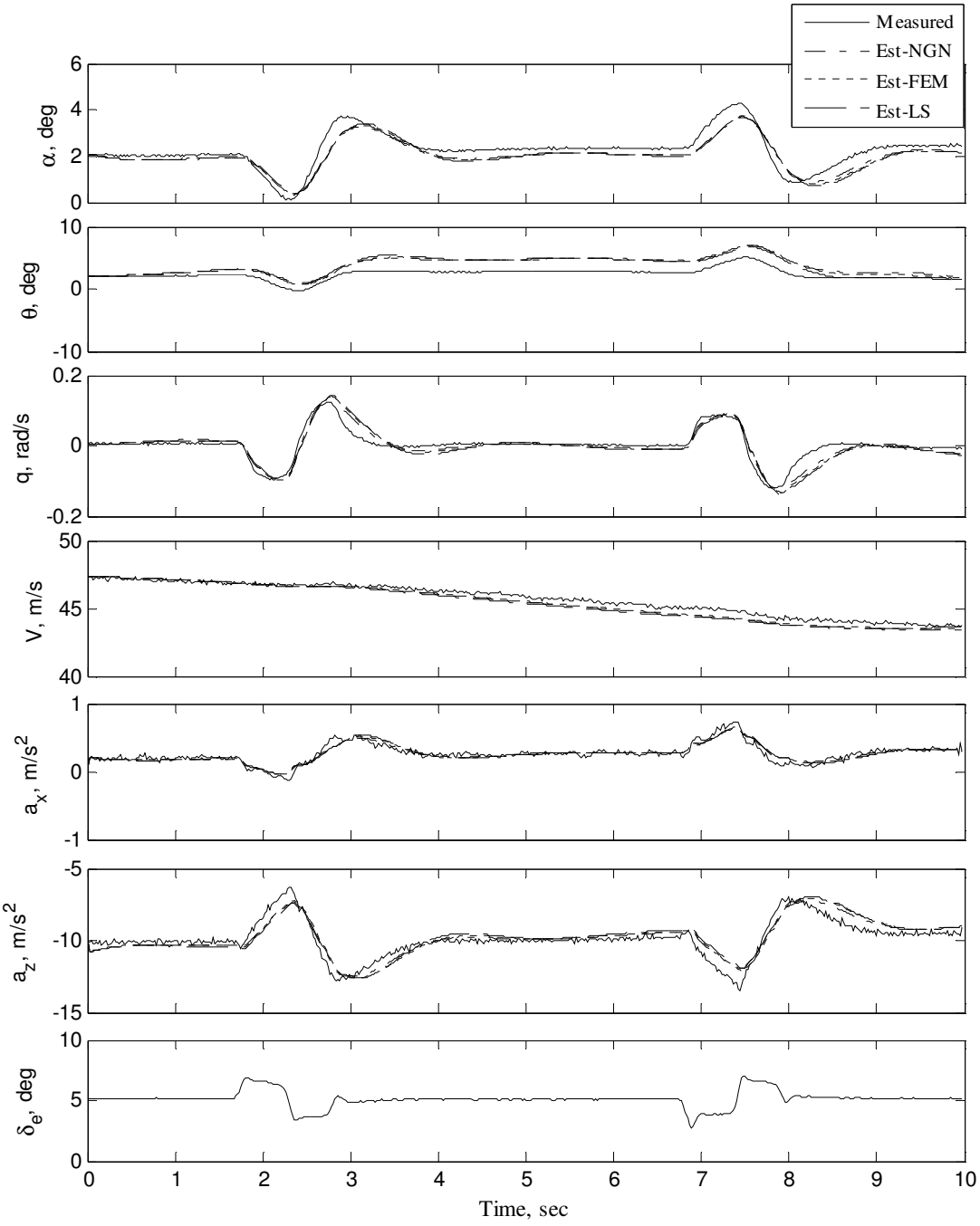


Figure 4. Model Verification: Proof-of-Match, FLT2

B. Parameter Estimated from Lateral-directional Data of ATTAS Aircraft using NGN, FEM, LS Method

The applicability of the NGN method was also investigated using real flight data pertaining to lateral-directional dynamics of ATTAS aircraft. The flight data FLT3 is presented in Fig. 3. The estimation of lateral-directional parameters in some sense was more involved as the flight data contained effects of two control inputs namely aileron, rudder and large number of parameters (as compared to longitudinal case) needed to be estimated.
The aerodynamic model pertaining to lateral-directional dynamics was assumed to have the following form.

$$C_Y = C_{Y_0} + C_{Y_\beta}\beta + C_{Y_p}\left(pb/2V\right) + C_{Y_r}\left(rb/2V\right) + C_{Y_{\delta r}}\delta r \tag{22}$$

$$C_\ell = C_{\ell_0} + C_{\ell_\beta}\beta + C_{\ell_p}\left(pb/2V\right) + C_{\ell_r}\left(rb/2V\right) + C_{\ell_{\delta a}}\delta a + C_{\ell_{\delta r}}\delta r \tag{23}$$

$$C_n = C_{n_0} + C_{n_\beta}\beta + C_{n_{\dot\beta}}\left(\dot\beta b/2V\right) + C_{n_p}\left(pb/2V\right) + C_{n_r}\left(rb/2V\right) + C_{n_{\delta r}}\delta r \tag{24}$$

For lateral-directional case, the parameters vector Θ to be estimated is given in Eq. (25).

$$\Theta = \left[\, C_{Y_0}\ C_{Y_\beta}\ C_{Y_p}\ C_{Y_r}\ C_{Y_{\delta_r}}\ C_{\ell_0}\ C_{\ell_\beta}\ C_{\ell_p}\ C_{\ell_r}\ C_{\ell_{\delta a}}\ C_{\ell_{\delta r}}\ C_{n_0}\ C_{n_\beta}\ C_{n_{\dot\beta}}\ C_{n_p}\ C_{n_r}\ C_{n_{\delta r}}\,\right]^T \tag{25}$$

The input vector $U(k)$ and $Z(k+1)$ for lateral-directional case were constructed as given in Eqs. (26) and (27) respectively.

$$U(k) = \left[\, C_Y(k)\ C_\ell(k)\ C_n(k)\ \beta(k)\ \phi(k)\ \psi(k)\ p(k)\ r(k)\,\right]^T \tag{26}$$

$$Z(k+1) = \left[\, \beta(k+1)\ \phi(k+1)\ \psi(k+1)\ p(k+1)\ r(k+1)\ a_y(k+1)\,\right]^T \tag{27}$$

The flight determined values of $C_Y(k), C_l(k)$ and $C_n(k)$ at the kth instant were obtained through the following equations.

$$C_Y(k) = ma_y^{CG} \big/ \bar{q}S \tag{28}$$

$$C_\ell^{CG}(k) = \left[\, I_x\dot p - I_{xz}\dot r - I_{xz}pq - (I_y - I_z)qr - \Delta F_{eng}\sin\sigma_{eng}Y_{ENCG}\,\right] \big/ \bar{q}Sb \tag{29}$$

$$C_n^{CG} = \left[\, I_z\dot r - I_{xz}\dot p + I_{xz}qr - (I_x - I_y)pq - \Delta F_{eng}\cos\sigma_{eng}Y_{ENCG}\,\right] \big/ \bar{q}Sb \tag{30}$$

Neural model representing the lateral-directional dynamics was arrived at using input vector $U(k)$ and $Z(k+1)$ as output vector. The parameter estimation process was initiated with some initial guess values. The parameters estimates along with their Cramer-Rao bounds obtained by applying NGN method on flight data FLT3 and FLT4 are presented in Table 2. It also lists the estimated values obtained using the maximum likelihood method,[20] and these estimated values of lateral-directional parameters are treated as reference values of the parameters of ATTAS aircraft. Further, estimates obtained after applying LS method are also presented in Table 2. Fairly good matching between the estimated values of the parameters with the reference values,[20] validates the applicability of the NGN method in estimating lateral-directional aerodynamic parameters from flight data. The estimated parameters namely, $C_{\ell_p}, C_{\ell_r}, C_{\ell_{\delta a}}, C_{\ell_{\delta r}}, C_{n_\beta}, C_{n_{\dot\beta}}$ and $C_{n_{\delta r}}$ were better estimated by NGN method as cooperated to estimate obtained by LS method (column 4 and 6). It may also be observed that the estimates have lower values of Cramer-Rao bound suggesting higher confidence in the estimates.

C. Parameter Estimated from Longitudinal Flight Data of HFB-320 with Atmospheric Turbulence using NGN, FEM, LS Method

All the three methods were applied on typical flight data having moderate turbulence level.[3] The estimated values of the parameters along with Cramer-Rao bound are listed in Table 3. It is observed that the estimates obtained using NGN method lies in close proximity with those obtained using FEM or LS method.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

Table 2. ATTAS Lateral Parameters

Parameter	Ref. 20	FLT3		FLT4	
		NGN	LS	NGN	LS
C_{Y_0}	-0.0035	-0.0087 (0.00012)*	-0.0067 (0.00005)	-0.0091 (0.00009)	-0.0069 (0.00005)
C_{Y_β}	-1.1562 (0.69)+	-1.1699 (0.00466)	-1.0487 (0.00218)	-1.1569 (0.00306)	-1.0346 (0.00159)
C_{Y_p}	0.2075 (8.12)	0.1396 (0.00827)	0.1736 (0.00391)	0.2605 (0.00756)	0.1451 (0.00396)
C_{Y_r}	0.5110 (6.96)	2.1034 (0.03616)	0.6408 (0.0164)	2.0399 (0.02174)	0.6121 (0.01111)
$C_{Y_{\delta_r}}$	0.2225 (3.79)	0.2518 (0.0072)	0.1924 (0.0035)	0.2577 (0.00463)	0.1933 (0.0024)
C_{ℓ_0}	0.00069	0.0002 (0.00008)	0.0001 (0.00002)	0.0004 (0.00007)	0.0003 (0.00002)
C_{ℓ_β}	-0.1164 (0.91)	-0.1329 (0.00335)	-0.1241 (0.00092)	-0.1385 (0.00233)	-0.1285 (0.00073)
C_{ℓ_p}	-1.0139 (0.22)	-0.9423 (0.01005)	-0.755 (0.0025)	-0.9059 (0.01012)	-0.7838 (0.00294)
C_{ℓ_r}	0.3195 (1.16)	0.3548 (0.02586)	0.2888 (0.00691)	0.3612 (0.01624)	0.2358 (0.00504)
$C_{\ell_{\delta a}}$	-0.2562 (0.20)	-0.2255 (0.00186)	-0.1929 (0.00047)	-0.2215 (0.00199)	-0.1975 (0.00058)
$C_{\ell_{\delta r}}$	0.0435 (2.16)	0.0457 (0.00546)	0.0455 (0.00148)	0.0398 (0.00357)	0.0427 (0.00109)
C_{n_0}	0.00135	0.0025 (0.00006)	0.0026 (0.00002)	0.0027 (0.00006)	0.0028 (0.00002)
C_{n_β}	0.2967 (0.11)	0.2566 (0.0019)	0.254 (0.0009)	0.2759 (0.00208)	0.2633 (0.0007)
$C_{n_{\dot\beta}}$	-0.2839 (1.54)	-0.2011 (0.02878)	-0.0492 (0.00226)	-0.287 (0.03409)	-0.0279 (0.00187)
C_{n_p}	-0.0489 (1.20)	-0.0395 (0.00329)	-0.0156 (0.0021)	-0.0506 (0.00458)	-0.0305 (0.00195)
C_{n_r}	-0.5996 (0.94)	-0.5782 (0.03012)	-0.6834 (0.02562)	-0.7504 (0.0350)	-0.4589 (0.02194)
$C_{n_{\delta r}}$	-0.1691 (0.36)	-0.1314 (0.00321)	-0.1424 (0.00143)	-0.1475 (0.0031)	-0.1474 (0.00098)

+ *Values in parentheses indicate standard deviation in %*

* *Values in parentheses indicate sample standard deviation.*

Table 3. HFB-320 Longitudinal Parameters

Parameter	NGN	FEM	LS
C_{D_0}	0.0567 (0.0008)*	0.0565 (0.00142)	0.0557 (0.00096)
C_{D_α}	0.3269 (0.00771)	0.3257 (0.01334)	0.3335 (0.00895)
C_{L_0}	0.0424 (0.00388)	0.0766 (0.00871)	0.0608 (0.00448)
C_{L_α}	4.6172 (0.0393)	4.2453 (0.08403)	4.4254 (0.04361)
C_{L_q}	113.362 (3.56)	93.9909 (5.87162)	105.406 (3.59418)
$C_{L_{\delta_e}}$	1.6059 (0.10127)	0.9792 (0.15853)	1.4377 (0.0976)
C_{m_0}	0.1299 (0.0039)	0.1159 (0.00164)	0.1193 (0.00189)
C_{m_α}	-1.0881 (0.039)	-0.9774 (0.01594)	-0.9837 (0.01844)
C_{m_q}	-46.582 (3.2105)	-42.826 (1.26458)	-44.051 (1.52001)
$C_{m_{\delta_e}}$	-1.668 (0.09844)	-1.6165 (0.03446)	-1.6253 (0.04127)

* *Values in parentheses indicate sample standard deviation.*

V. Conclusions

A new method christened the Neural-Gauss-Newton (NGN) method has been proposed for estimating aircraft parameters from flight data using feed forward neural networks. The proposed method advantageously uses the universal function mapping characteristics of feed forward neural network and optimization capability of Gauss-Newton algorithm. The results obtained for real flight data pertaining to longitudinal and lateral-directional dynamics have shown the success and the potential of the proposed method. Since the proposed method does not require solving of equations of motion, it presents itself as straight forward method in which the FFNN is trained to capture the flight dynamic model of an aircraft, and the parameters along with Cramer-Rao bounds are estimated in

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941

few iterations. The NGN method, which bypasses the requirement of solving the equation of motion, may have special significance in handling flight data of unstable airplane.

References

[1]Klein, V., and Morelli, E. A., *Aircraft System Identification – Theory and Practice*, AIAA Educaton Series, Inc., Reston, Virginia, 2006.

[2]Maine, K. E., and Iliff, K. W., "Application of Parameter Estimation to Aircraft Stability and Control: The Output Error Approach," NASA RP 1168, Jan. 1986.

[3]Jategaonkar, R. V., *Flight Vehicle System Identification – A Time Domain Methodology*, AIAA, Inc., Reston, VA, 2006.

[4]Iliff, K. W., "Parameter Estimation for Flight Vehicle," *Journal of Guidance, Control and Dynamics*, Vol. 12, No. 5, 1989, pp. 609-622.

[5]Hamel, P. G., and Jategaonkar, R. V., "The Evolution of Flight Vehicle System Identification," AGARD, DLR Germany, 8-10, May 1995.

[6]Balakrishnan, A. V., *Stochastic System Identification Techniques*, in Stochastic Optimization and control, edited by H. F. Karreman, John Wiley & Sons, London, 1968.

[7]Mehra, R. K., "Maximum Likelihood Identification of Aircraft Parameters," Proceedings of the 11th Joint Automatic Control Conference, Atlanta, GA, 1970, pp. 442-444.

[8]Mehra, R. K., "Identification of Stochastic Linear Dynamic Systems Using Kalman Filter Representations," *AIAA Journal*, Vol. 9, Jan. 1971, pp. 28-31.

[9]Jategaonkar, R. V. and Plaetschke, E., "Estimation of Aircraft Parameters Using Filter Error Methods and Extended Kalman Filter," DFVLR-FB 88-15, March 1988.

[10]Raisinghani, S. C., Ghosh, A. K. and Kalra, P. K., "Two New Techniques for Aircraft Parameter Estimation Using Neural Networks," *The Aeronautical Journal*, Vol. 102, No. 1011, Jan. 1998, pp. 25-29.

[11]Ghosh, A. K., and Raisinghani, S. C., "Parameter Estimation from Flight Data of an Unstable Aircraft using Neural Networks," *Journal of Aircraft*, Vol. 35, No. 5, 2002, pp. 892-894.

[12]Hornik, K., Stinchcombe, M. and White, H., "Multi Layer Feed Forward Neural Networks are Universal Approximators," *Neural Networks*, Vol. 2, No. 5, 1989, pp. 359-366.

[13]Cichocki, A. and Unbehauen, R., *Neural Networks for Optimization and Signal Processing*, John Wiley & Sons, New York, 1993.

[14]Hess, R. A., "On the Use of Back Propagation with Feed Forward Neural Networks for the Aerodynamic Estimation Problem," *AIAA Paper* 93-3639, Aug. 1993.

[15]Linse, D. J. and Stengel, R. F., "Identification of Aerodynamic Coefficients using Computational Neural Networks," *Journal of Guidance, Control and Dynamics*, Vol. 16, No. 6, Nov.-Dec. 1993, pp. 1018-1025.

[16]Youseff, H. M., and Jaung, J. C., "Estimation of Aerodynamic Coefficients using Neural Networks," *AIAA Paper* 93-3639, Aug. 1993.

[17]Peyada, N. K. and Ghosh, A. K., "Aerodynamic Parameter Estimation using New Filtering Technique Based on Neural Network and Gauss-Newton Method ", *ARMS 2008,* ARDE, Pashan, Pune, March 28-29, 2008.

[18]Rangaranjan, R. and Viswanathan, S., "Wind tunnel test results on a 1/5 scale HANSA model," NAL TR-01, Vol. 1, Dec. 1997.

[19]Ghosh, A.K., Raisinghani, S.C., and Khubchandani, S., "Estimation of Aircraft Lateral-Directional Parameters using Neural Networks," *Journal of Aircraft*, Vol. 35, No. 6, Nov.-Dec. 1998, pp. 876-881.

[20]Jategaonkar, R. V., "Identification of the Aerodynamic Model of the DLR Research Aircraft ATTAS from Flight Test Data," DLR, DLR-FB 90-40, Braunschweig, Germany, September 1990.

Downloaded by INST TEC DE AERONAUTICA (ITA) on May 3, 2025 | http://arc.aiaa.org | DOI: 10.2514/6.2009-5941